
Use Cases

for

FamPlan

Version 1.0 - Lab 1 Submission

Prepared by Banana Five-O

NTU SC2006 Software Engineering

8 September 2026

Revision History

Name	Date	Reason For Changes	Version
Banana Five-O	8 September 2026	Initial use-case report for Lab 1 submission	1.0

Actor and Relationship Model

Actor or Relationship	Meaning in FamPlan
Visitor	Base public actor for search, profile, market, location, comparison, and indicative HFE functions.
Registered User	Inherits Visitor behaviour and adds account, preference, shortlist, and private-note functions.
Data Administrator	Inherits Registered User behaviour and adds source-status and manual-refresh functions.
System (scheduled process)	Initiates UC-8.3 without a user present.
Supporting systems	Email Delivery Service supports UC-1.3; OneMap supports UC-5.1 and UC-5.2; Configured Data Sources support UC-8.2 and UC-8.3.
Relationships	UC-3.2 extends UC-3.1. UC-4.1 and UC-6.2 include UC-6.1. UC-6.1 has no independent initiating actor.

Team Contributions and AI Use

Member	Username	Team Leader	Lab 1 Contribution
POLOYAPOY RON JOSHUA ABES	POLO0002	Yes	Data Dictionary
LIM RUI XIAN EDWARD	ELIM095	No	Functional and Non-functional Requirements
LE VIET HOANG ANH	HOANGANH003	No	Initial Use Case Diagram
ATHENA LIM HUI RU (LIN HUIRU)	ALIM102	No	Initial Use Case Model
MUHAMMAD ILHAM BIN IRWAN	MUHA0313	No	UI Mockups

AI-use disclosure: OpenAI Codex assisted with consolidating team-authored artifacts, consistency editing, and populating the supplied Word templates; the team reviewed and accepts responsibility for the final content.

Use Case 1.1 Register Account

Use Case ID:	1.1		
Use Case Name:	Register Account		
Created By:	Banana Five-O	Last Updated By:	Banana Five-O
Date Created:	8 September 2026	Date Last Updated:	8 September 2026
Actor:	Visitor		
Description:	Allows an unauthenticated visitor to create a new user account with validated credentials and consent to data terms. Trigger: The Visitor selects Create Account.		
Preconditions:	1. Visitor is not authenticated. 2. Registration page is available.		
Postconditions:	Success: Account created; visitor may log in. Failure: No account created; field-specific errors shown.		
Priority:	Must		
Frequency of Use:	Low (once per user)		
Flow of Events:	1. Visitor opens the registration page. 2. Visitor enters display name, email, password, and data-use consent. 3. Visitor selects "Create Account." 4. System validates the submitted credentials, password strength, and mandatory data-use consent. 5. System creates the account record with a hashed password (NFR-SEC-02). 6. System displays a registration-confirmed screen and prompts the visitor to log in.		
Alternative Flows:	1.1.AC.1: Duplicate email -> field error "This email is already registered"; return to step 2. 1.1.AC.2: Malformed email -> field error "Enter a valid email address"; return to step 2. 1.1.AC.3: Weak password -> field error naming the unmet rule; return to step 2. 1.1.AC.4: Missing consent -> field error "You must accept the data-use terms to continue"; return to step 2.		
Exceptions:	1.1.EX.1: System unavailable -> generic error shown; no account created.		
Includes:	None.		
Special Requirements:	Functional requirements: FR-ACC-01. Additional constraints: NFR-SEC-02 (salted-hash password storage); NFR-USAB-02 (every rejected field names itself and how to fix it); NFR-PRIV-03 (no personal data collected beyond name, email, password, consent).		
Assumptions:	Email address is sufficient as a unique identifier; no phone verification or SSO is required for MVP.		
Notes and Issues:	Do not collect personal data beyond authentication and declared course needs.		

Use Case 1.2 Login / Logout

Use Case ID:	1.2		
Use Case Name:	Login / Logout		
Created By:	Banana Five-O	Last Updated By:	Banana Five-O
Date Created:	8 September 2026	Date Last Updated:	8 September 2026
Actor:	Registered User		
Description:	Allows a Registered User holding a registered account to authenticate into the system, and to end an authenticated session. Trigger: The Registered User selects Log In or, from an authenticated session, Log Out.		
Preconditions:	1. Registered User holds a registered account (for login). 2. Registered User is authenticated (for logout).		
Postconditions:	Success (login): Authenticated session established. Success (logout): Session invalidated; Registered User returned to a public page. Failure: Generic authentication failure shown; no session created.		
Priority:	Must		
Frequency of Use:	High (every visit)		
Flow of Events:	1. Registered User navigates to the login page. 2. Registered User enters email and password. 3. System validates the credentials against the stored hash. 4. System establishes an authenticated session token and redirects the Registered User to the home/search page. 5. Registered User later selects "Log out." 6. System invalidates the session token server-side. 7. System returns the Registered User to a public page.		
Alternative Flows:	1.2.AC.1: Incorrect password -> generic authentication failure "Email and password do not match"; return to step 2; failed-attempt counter incremented (NFR-SEC-04). 1.2.AC.2: Unknown account -> identical generic failure shown, to prevent account enumeration. 1.2.AC.3: Account locked after five consecutive failures -> generic failure shown; lockout lasts 15 minutes; counter resets on next successful login.		
Exceptions:	1.2.EX.1: System unavailable -> generic error; no session created.		
Includes:	None.		
Special Requirements:	Functional requirements: FR-ACC-02, FR-ACC-03. Additional constraints: NFR-SEC-04 (5-attempt lockout, 15-minute duration, counter resets on success); NFR-SEC-01 (HTTPS/TLS 1.2+ only, plain HTTP redirects without accepting credentials).		
Assumptions:	Email and password is the sole authentication mechanism for MVP (no SSO/OAuth).		
Notes and Issues:	Protected pages and APIs must reject the old session token immediately after logout - verify no stale-token race condition.		

Use Case 1.3 Reset Password

Use Case ID:	1.3		
Use Case Name:	Reset Password		
Created By:	Banana Five-O	Last Updated By:	Banana Five-O
Date Created:	8 September 2026	Date Last Updated:	8 September 2026
Actor:	Registered User		
Description:	Allows a Registered User who has forgotten their password to reset it through a time-limited, single-use link sent to their registered email. Trigger: The Registered User selects Forgot Password.		
Preconditions:	Registered User holds a registered account with a valid email address on file.		
Postconditions:	Success: Password updated; Registered User can log in with the new password. Failure: No password changed; a generic message is shown that does not reveal whether the account exists.		
Priority:	Should		
Frequency of Use:	Low		
Flow of Events:	1. Registered User selects Forgot Password and enters the account email address. 2. System accepts the request and displays the same generic confirmation for every submitted address. 3. For a registered address, System creates a time-limited, single-use reset token and asks Email Delivery Service to deliver the reset link. 4. Registered User opens the valid reset link and enters a new password twice. 5. System validates the token and the new password against the password policy. 6. System stores the new password hash and invalidates the reset token and existing sessions. 7. System confirms the change and directs the Registered User to log in.		
Alternative Flows:	1.3.AC.1: Unknown email -> same generic confirmation shown (step 4). 1.3.AC.2: Expired token -> error "This link has expired, request a new one"; Registered User must restart from step 1. 1.3.AC.3: Reused (already-consumed) token -> rejected with an error; Registered User must restart from step 1.		
Exceptions:	1.3.EX.1: Email delivery failure -> no distinguishing error shown to the Registered User, to avoid account enumeration; failure logged server-side for the Data Administrator.		
Includes:	None.		
Special Requirements:	Functional requirements: FR-ACC-04. Additional constraints: NFR-SEC-02 (passwords stored only as salted hashes, never logged).		
Assumptions:	Registered User has access to the inbox of their registered email address.		
Notes and Issues:	None.		

Use Case 1.4 Manage Account Details

Use Case ID:	1.4		
Use Case Name:	Manage Account Details		
Created By:	Banana Five-O	Last Updated By:	Banana Five-O
Date Created:	8 September 2026	Date Last Updated:	8 September 2026
Actor:	Registered User		
Description:	Allows a Registered User to update the account display name; email and credentials are managed separately. Trigger: The Registered User opens account details and submits a new display name.		
Preconditions:	1. Registered User is authenticated. 2. The account is active.		
Postconditions:	Success: Updated account record persisted; confirmation shown. Failure: Validation errors shown; no changes persisted.		
Priority:	Must		
Frequency of Use:	Low		
Flow of Events:	1. Registered User opens account details. 2. System displays the current display name and identifies it as editable. 3. Registered User enters a new display name and submits the change. 4. System validates the value. 5. System saves the new display name. 6. System displays confirmation and the updated account details.		
Alternative Flows:	1.4.AC.1: Invalid field value -> field-specific error; return to step 3. 1.4.AC.2: Attempt to edit a disallowed/sensitive field (password, authentication email) -> rejected with an explanation directing the Registered User to the dedicated flow.		
Exceptions:	1.4.EX.1: System unavailable -> generic error; no changes persisted.		
Includes:	None.		
Special Requirements:	Functional requirements: FR-ACC-05. Additional constraints: NFR-USAB-02 (every rejected input names the affected field and how to correct it).		
Assumptions:	Sensitive fields (e.g. password, authentication email) are changed through dedicated flows (1.3 Reset Password), not this general edit form.		
Notes and Issues:	None.		

Use Case 1.5 Delete Account

Use Case ID:	1.5		
Use Case Name:	Delete Account		
Created By:	Banana Five-O	Last Updated By:	Banana Five-O
Date Created:	8 September 2026	Date Last Updated:	8 September 2026
Actor:	Registered User		
Description:	Allows a Registered User to permanently delete their own account, saved housing preferences, shortlist, and private notes, after explicit confirmation. Trigger: The Registered User requests account deletion and confirms the action.		
Preconditions:	Registered User is authenticated.		
Postconditions:	Success: Account, saved preferences, shortlist entries, and notes are irretrievable through the application within 24 hours; Registered User is logged out. Failure: No data removed; account remains active.		
Priority:	Must		
Frequency of Use:	Low		
Flow of Events:	1. Registered User navigates to account settings and selects "Delete account." 2. System displays a warning describing what will be deleted (account, saved preferences, shortlist, notes) and that the action is irreversible. 3. Registered User confirms deletion via an explicit confirmation prompt. 4. System marks the account, saved preferences, shortlist, and notes for deletion. 5. System invalidates the current session and logs the Registered User out. 6. System permanently erases the marked records within 24 hours (NFR-PRIV-02).		
Alternative Flows:	1.5.AC.1: Registered User cancels confirmation prompt -> deletion aborted; return to account settings.		
Exceptions:	1.5.EX.1: System unavailable at step 4 -> generic error; no records marked for deletion.		
Includes:	None.		
Special Requirements:	Functional requirements: FR-ACC-06. Additional constraints: NFR-PRIV-02 (irretrievable through the application within 24 hours of confirmed deletion, verified at UI, API, and direct authorised data-store level).		
Assumptions:	Deletion is self-service only; no administrator override or recovery window is provided for MVP.		
Notes and Issues:	None.		

Use Case 2.1 Manage Housing Preferences

Use Case ID:	2.1		
Use Case Name:	Manage Housing Preferences		
Created By:	Banana Five-O	Last Updated By:	Banana Five-O
Date Created:	8 September 2026	Date Last Updated:	8 September 2026
Actor:	Registered User		
Description:	Allows a Registered User to store and update their budget range, preferred towns/flat types/thresholds, priority commute destinations, amenity and accessibility preferences, and criterion importance weights. Trigger: The Registered User opens or saves housing preferences.		
Preconditions:	Registered User is authenticated.		
Postconditions:	Success: Preferences saved and available to apply to searches (3.2) and to scoring (6.1). Failure: Field-specific validation errors shown; no changes persisted.		
Priority:	Must		
Frequency of Use:	Low to medium (set once, revisited occasionally)		
Flow of Events:	1. Registered User opens housing preferences. 2. System displays the saved values, or documented defaults when no values exist. 3. Registered User sets budget, towns, flat types, minimum floor area, and minimum remaining lease. 4. Registered User selects up to three commute destinations, which System resolves through OneMap. 5. Registered User selects amenity categories, barrier-free access preference, and a 0-5 importance weight for each scoring criterion. 6. Registered User submits the preferences and System validates all values. 7. System saves the validated preferences and displays confirmation.		
Alternative Flows:	2.1.AC.1: Minimum budget greater than maximum -> field error "Minimum budget cannot exceed maximum." 2.1.AC.2: Non-numeric or negative threshold -> field error naming the affected field. 2.1.AC.3: Fourth or duplicate destination selected -> rejected with an explanation; step 4 repeated. 2.1.AC.4: Unresolvable destination -> field error "Could not locate this destination." 2.1.AC.5: All seven weights set to zero -> accepted; system applies equal-weight fallback (see 6.1).		
Exceptions:	2.1.EX.1: System unavailable -> generic error; no changes persisted.		
Includes:	None.		
Special Requirements:	Functional requirements: FR-PREF-01 to FR-PREF-06. Additional constraints: NFR-PRIV-03 (no personal data collected beyond what preferences require).		
Assumptions:	A saved preference set is scoped to one Registered User account.		
Notes and Issues:	None.		

Use Case 3.1 Search and Filter Flats

Use Case ID:	3.1		
Use Case Name:	Search and Filter Flats		
Created By:	Banana Five-O	Last Updated By:	Banana Five-O
Date Created:	8 September 2026	Date Last Updated:	8 September 2026
Actor:	Visitor (Registered User inherits this use case)		
Description:	Allows any Visitor to search HDB resale block-and-flat profiles by town, street, or block, and to filter and sort the results. Trigger: The Visitor submits search text, filters, or a sort selection.		
Preconditions:	None - available to unauthenticated visitors.		
Postconditions:	Success: Matching profiles displayed, labelled as block-and-flat profiles, in the selected order. Failure/empty: No-results state or field-specific rejection shown.		
Priority:	Must		
Frequency of Use:	High		
Flow of Events:	1. Visitor enters search text or chooses to browse available flat profiles. 2. Visitor selects any price, flat type, town, floor-area, remaining-lease, and sort options. 3. System validates the search and filter values. 4. System searches the local dataset using the validated criteria. 5. System orders the matching profiles using the selected sort rule and places unavailable sort values last. 6. System displays block-and-flat profile results with the active criteria. 7. Visitor selects a result or revises the search criteria.		
Alternative Flows:	3.1.AC.1: Empty search query with no filters -> returns default paginated view of all available flat profiles. 3.1.AC.2: Contradictory filter combination -> field-specific rejection before execution; no request run; return to step 2. 3.1.AC.3: Unknown query term -> no-results state shown at step 7.		
Exceptions:	3.1.EX.1: Underlying data store unavailable -> controlled error/loading state shown; no partial results.		
Includes:	None.		
Special Requirements:	Functional requirements: FR-SRCH-01, FR-SRCH-02, FR-SRCH-04 to FR-SRCH-06. Additional constraints: NFR-PERF-01 (95th-percentile response <=2s at 100 concurrent users against >=1,000,000 records).		
Assumptions:	Results represent historical profiles, not live listings.		
Notes and Issues:	None.		

Use Case 3.2 Apply Saved Preferences

Use Case ID:	3.2		
Use Case Name:	Apply Saved Preferences		
Created By:	Banana Five-O	Last Updated By:	Banana Five-O
Date Created:	8 September 2026	Date Last Updated:	8 September 2026
Actor:	Registered User		
Description:	Allows a Registered User to populate the current search's filter criteria from their saved housing preferences. Trigger: During UC-3.1, the Registered User selects Apply Saved Preferences.		
Preconditions:	Registered User is authenticated and has at least one saved preference value (see 2.1).		
Postconditions:	Success: Search filters populated from the saved preference set. Failure: No change to the current filters.		
Priority:	Must		
Frequency of Use:	Medium		
Flow of Events:	1. Registered User is on the search page (3.1 Search and Filter Flats). 2. Registered User selects "Apply saved preferences." 3. System retrieves the Registered User's saved preference values. 4. System populates the search filters (budget, towns, flat types, thresholds) with the retrieved values. 5. System re-executes the search with the applied filters.		
Alternative Flows:	3.2.AC.1: No saved preferences exist -> action disabled or explanatory message shown; no filters changed.		
Exceptions:	3.2.EX.1: System unavailable -> generic error; filters unchanged.		
Includes:	Extends UC-3.1 Search and Filter Flats at the saved-preferences extension point.		
Special Requirements:	Functional requirements: FR-SRCH-03. Additional constraints: None		
Assumptions:	Saved preference values are already validated at save time (see 2.1).		
Notes and Issues:	None.		

Use Case 4.1 View Flat Profile Details

Use Case ID:	4.1		
Use Case Name:	View Flat Profile Details		
Created By:	Banana Five-O	Last Updated By:	Banana Five-O
Date Created:	8 September 2026	Date Last Updated:	8 September 2026
Actor:	Visitor (Registered User inherits this use case)		
Description:	Displays the available block and flat-profile attributes for a profile selected from search results, together with its suitability score. Trigger: The Visitor selects a block-and-flat profile.		
Preconditions:	A block-and-flat profile has been selected (see 3.1).		
Postconditions:	Success: Attributes and score displayed with source and date. Failure: Explicit unavailable-attribute state shown for missing fields.		
Priority:	Must		
Frequency of Use:	High		
Flow of Events:	1. Visitor selects a profile from search results. 2. System retrieves the block and flat-profile record. 3. System displays address, town, flat type, floor area, lease commencement year, source provider, last update date, and estimated remaining lease (calculated within a 12-month approximation). 4. System calculates and displays the suitability score for the profile using the included use case 6.1 Calculate Suitability Score.		
Alternative Flows:	4.1.AC.1: An attribute is unavailable from source -> explicit "not available" state shown for that field, not a blank or zero.		
Exceptions:	4.1.EX.1: Source data unreachable -> stale-data or unavailable notice shown for the cached record.		
Includes:	UC-6.1 Calculate Suitability Score.		
Special Requirements:	Functional requirements: FR-MKT-01; FR-DEC-01 to FR-DEC-03 through included UC-6.1. Additional constraints: NFR-DATA-01 (source and date shown for every output); NFR-DATA-02 (stale-data marking beyond 31 days).		
Assumptions:	Remaining lease is accurate only to within twelve months, since lease commencement is published as a year.		
Notes and Issues:	None.		

Use Case 4.2 View Historical Market Trends

Use Case ID:	4.2		
Use Case Name:	View Historical Market Trends		
Created By:	Banana Five-O	Last Updated By:	Banana Five-O
Date Created:	8 September 2026	Date Last Updated:	8 September 2026
Actor:	Visitor (Registered User inherits this use case)		
Description:	Displays comparable historical resale transactions and aggregate statistics for a selected block and flat type over a chosen period. Trigger: The Visitor opens historical market trends for a selected profile.		
Preconditions:	A block-and-flat profile has been selected (see 4.1).		
Postconditions:	Success: Transaction set, aggregates, and trend visualisation displayed with source and date. Failure: Explicit no-data, insufficient-data, or source-unavailable state shown.		
Priority:	Must		
Frequency of Use:	High		
Flow of Events:	1. Visitor opens historical market analysis for the selected profile. 2. Visitor selects a 12-, 36-, or 60-month analysis period. 3. System retrieves at least five transactions matching the block and flat type for that period. 4. System calculates transaction count, median price, minimum price, and maximum price. 5. System renders a time-series chart and an accessible summary data table. 6. System displays the source, coverage period, last-update date, and indicative-evidence label.		
Alternative Flows:	4.2.AC.1: No matching transactions are available; System shows a no-data state and suppresses the aggregates. 4.2.AC.2: One to four matching transactions are available; System labels the history as insufficient and suppresses unsupported aggregates.		
Exceptions:	4.2.EX.1: Source unavailable -> explicit source-unavailability state, distinct from a zero value.		
Includes:	None.		
Special Requirements:	Functional requirements: FR-MKT-02 to FR-MKT-08. Additional constraints: NFR-DATA-01, NFR-DATA-02.		
Assumptions:	Published transactions are sufficient evidence; no price prediction is made.		
Notes and Issues:	None.		

Use Case 5.1 Explore Map and Amenities

Use Case ID:	5.1		
Use Case Name:	Explore Map and Amenities		
Created By:	Banana Five-O	Last Updated By:	Banana Five-O
Date Created:	8 September 2026	Date Last Updated:	8 September 2026
Actor:	Visitor (Registered User inherits this use case)		
Description:	Displays a selected block on an interactive map, along with nearby MRT/bus stations, schools, and amenities within a chosen radius, with calculated distances. Trigger: The Visitor opens the map and nearby-amenities view for a selected profile.		
Preconditions:	A block-and-flat profile with coordinates has been selected.		
Postconditions:	Success: Map and categorised nearby-place results displayed with distances, provider, and retrieval time. Failure: Category-specific unavailable state shown.		
Priority:	Must		
Frequency of Use:	High		
Flow of Events:	1. Visitor opens the location view for the selected profile. 2. System displays an interactive map centred on the block with a labelled marker. 3. Visitor selects one or more MRT/bus, school, or amenity categories. 4. Visitor selects a radius of 500 m, 1 km, or 2 km. 5. System requests nearby places for the selected categories and radius from OneMap. 6. System calculates and displays the distance from the block to each returned place. 7. System displays the provider name and retrieval time.		
Alternative Flows:	5.1.AC.1: No results in a category within the radius -> explicit empty-category state for that category only, others unaffected. 5.1.AC.2: Block coordinates missing -> map shows an explanatory state instead of a marker; categories/radius controls disabled.		
Exceptions:	5.1.EX.1: Location provider unavailable -> cached results shown with an age/staleness notice, or an explicit unavailable state; unrelated functions remain usable (FR-RES-02).		
Includes:	None.		
Special Requirements:	Functional requirements: FR-LOC-01 to FR-LOC-04, FR-LOC-07. Additional constraints: NFR-ACC-02 (keyboard operability with visible focus); NFR-REL-03 (dependency failure isolated from unrelated functions).		
Assumptions:	OneMap is the geocoding/nearby-place provider; schools are pre-geocoded (Assumption A-4).		
Notes and Issues:	None.		

Use Case 5.2 Check Commute Route

Use Case ID:	5.2		
Use Case Name:	Check Commute Route		
Created By:	Banana Five-O	Last Updated By:	Banana Five-O
Date Created:	8 September 2026	Date Last Updated:	8 September 2026
Actor:	Visitor (Registered User inherits this use case)		
Description:	Requests a route and estimated travel time from a selected block to a chosen destination for a supported travel mode, including barrier-free routing where requested. Trigger: The Visitor requests a route from a selected profile to a destination.		
Preconditions:	A block-and-flat profile has been selected; a destination is available (searched, or a saved priority destination from 2.1).		
Postconditions:	Success: Route, distance, duration, and retrieval time displayed. Failure: Explicit unavailable-route or unsupported-mode state shown.		
Priority:	Must		
Frequency of Use:	Medium		
Flow of Events:	1. Visitor selects a destination, travel mode, and any barrier-free routing preference. 2. System validates and resolves the destination. 3. System requests the route from the selected block to the destination through OneMap. 4. System displays the route, distance, and estimated duration. 5. System displays the provider and retrieval time for the route information.		
Alternative Flows:	5.2.AC.1: Barrier-free routing requested but not covered by the provider -> explicit coverage-unavailable statement shown instead of a route. 5.2.AC.2: Unsupported travel mode -> explanatory message; no request sent. 5.2.AC.3: Invalid or unresolved destination -> field-specific error; return to step 1.		
Exceptions:	5.2.EX.1: Routing provider timeout or no-route response -> explicit unavailable state; other functions remain usable.		
Includes:	None.		
Special Requirements:	Functional requirements: FR-LOC-05 to FR-LOC-07. Additional constraints: NFR-PERF-03 (95th-percentile end-to-end route response <=5s under normal provider conditions).		
Assumptions:	OneMap's barrier-free coverage is real but partial (Assumption A-3).		
Notes and Issues:	None.		

Use Case 6.1 Calculate Suitability Score

Use Case ID:	6.1		
Use Case Name:	Calculate Suitability Score		
Created By:	Banana Five-O	Last Updated By:	Banana Five-O
Date Created:	8 September 2026	Date Last Updated:	8 September 2026
Actor:	No initiating actor - included behaviour invoked by UC-4.1 or UC-6.2		
Description:	Calculates a suitability score for a profile from the active criterion weights, with an itemised, reconcilable explanation and explicit treatment of missing data. Trigger: UC-4.1 or UC-6.2 requests the included suitability-score behaviour.		
Preconditions:	1. A block-and-flat profile is available. 2. A scoring ruleset and criterion weights are available.		
Postconditions:	Success: Total score and itemised explanation displayed. Failure: N/A - a documented rule is always applied, including for missing data.		
Priority:	Must		
Frequency of Use:	High (invoked by 4.1, 6.2, and shortlist views)		
Flow of Events:	1. UC-4.1 or UC-6.2 requests a suitability score for a profile. 2. System retrieves the profile evidence and the active saved or default criterion weights. 3. System identifies unavailable criterion data and applies the documented missing-data treatment. 4. System calculates the score using the versioned scoring rules. 5. System rounds the total to the nearest whole number, with an exact 0.5 rounded upward. 6. System displays the 0-100 total and each criterion's value, weight, and contribution. 7. System displays the ruleset and data versions and confirms that the contributions reconcile with the total.		
Alternative Flows:	6.1.AC.1: A criterion lacks sufficient data -> system labels it as missing and applies the documented weight-adjustment rule, disclosed in the explanation. 6.1.AC.2: All seven weights are zero -> system applies equal weights (1.0 each) across all criteria and displays an on-screen notice explaining the equal-weight fallback.		
Exceptions:	6.1.EX.1: None - calculation is local with no external dependency.		
Includes:	Included by UC-4.1 and UC-6.2.		
Special Requirements:	Functional requirements: FR-DEC-01 to FR-DEC-03. Additional constraints: NFR-TRANS-01 (score fully reproducible from the displayed explanation); NFR-DATA-03 (identical inputs produce identical scores).		
Assumptions:	Normalisation constants and criterion weights are maintained as versioned configuration (NFR-MAINT-01).		
Notes and Issues:	None.		

Use Case 6.2 Compare Flat Profiles

Use Case ID:	6.2		
Use Case Name:	Compare Flat Profiles		
Created By:	Banana Five-O	Last Updated By:	Banana Five-O
Date Created:	8 September 2026	Date Last Updated:	8 September 2026
Actor:	Visitor (Registered User inherits this use case)		
Description:	Displays two to four selected profiles side by side with their attributes, market evidence, location measures, and score breakdowns. Trigger: The Visitor selects between two and four profiles and requests comparison.		
Preconditions:	At least two profiles have been selected for comparison.		
Postconditions:	Success: Side-by-side comparison view displayed. Failure: Selection-count or duplicate-selection error shown.		
Priority:	Must		
Frequency of Use:	High		
Flow of Events:	1. Visitor selects between two and four profiles for comparison. 2. For each selected profile, system retrieves the suitability score using the included use case 6.1 Calculate Suitability Score. 3. System retrieves attributes, market evidence, and location measures for each profile. 4. System displays all selected profiles side by side against a common set of criteria. 5. Visitor may remove a profile from the comparison or add another, within the two-to-four range; system re-renders the view after each change.		
Alternative Flows:	6.2.AC.1: Fewer than two or more than four profiles selected -> rejected with an explanation; selection unchanged. 6.2.AC.2: Duplicate profile selected -> rejected; no duplicate entry added. 6.2.AC.3: A profile is missing a value for a compared criterion -> explicit missing-data state shown for that cell.		
Exceptions:	6.2.EX.1: System unavailable -> generic error; no comparison view rendered.		
Includes:	UC-6.1 Calculate Suitability Score.		
Special Requirements:	Functional requirements: FR-DEC-04; FR-DEC-01 to FR-DEC-03 through included UC-6.1. Additional constraints: NFR-PERF-02 (95th-percentile response <=3s for two-, three-, and four-profile comparisons).		
Assumptions:	None		
Notes and Issues:	None.		

Use Case 6.3 Manage Shortlist and Notes

Use Case ID:	6.3		
Use Case Name:	Manage Shortlist and Notes		
Created By:	Banana Five-O	Last Updated By:	Banana Five-O
Date Created:	8 September 2026	Date Last Updated:	8 September 2026
Actor:	Registered User		
Description:	Allows a Registered User to add or remove profiles from a personal shortlist, and to create, edit, or clear a private note on each shortlisted profile. Trigger: The Registered User adds or removes a shortlist entry, or edits its private note.		
Preconditions:	Registered User is authenticated; a profile is selected for the add/remove action.		
Postconditions:	Success: Shortlist and/or note updated and persisted. Failure: Validation error shown; no change persisted.		
Priority:	Must (shortlist); Should (notes)		
Frequency of Use:	Medium		
Flow of Events:	1. Registered User opens a profile or the saved shortlist. 2. Registered User selects a shortlist action for a profile. 3. System verifies ownership, duplicate status, and the ten-profile capacity. 4. System adds or removes the profile and displays the updated shortlist. 5. For a saved profile, Registered User enters, edits, or clears a private note of up to 500 characters. 6. System validates and saves the note. 7. System displays confirmation and the persisted shortlist state.		
Alternative Flows:	6.3.AC.1: Profile already shortlisted -> "already saved" state shown; no duplicate created. 6.3.AC.2: Remove a non-existing entry -> no-op with confirmation. 6.3.AC.3: Note exceeds the length boundary or contains unsafe markup -> validation error; return to step 4. 6.3.AC.4: Access attempted from another account -> rejected.		
Exceptions:	6.3.EX.1: System unavailable -> generic error; no changes persisted.		
Includes:	None.		
Special Requirements:	Functional requirements: FR-DEC-05 to FR-DEC-07. Additional constraints: None		
Assumptions:	None		
Notes and Issues:	None.		

Use Case 7.1 Perform Indicative HFE Pre-check

Use Case ID:	7.1		
Use Case Name:	Perform Indicative HFE Pre-check		
Created By:	Banana Five-O	Last Updated By:	Banana Five-O
Date Created:	8 September 2026	Date Last Updated:	8 September 2026
Actor:	Visitor (Registered User inherits this use case)		
Description:	Walks a Visitor through a guided questionnaire to classify general HDB resale-purchase eligibility, grant categories, and loan readiness as indicative guidance only, without persisting any answers. Trigger: The Visitor starts the pre-check and acknowledges the indicative-use disclosure.		
Preconditions:	Visitor acknowledges the pre-check is general guidance, not an official HFE assessment.		
Postconditions:	Success: Classification with reasons, sources, and a link to HDB's official process displayed; session answers discarded. Failure: Pre-check not started, or stopped with field-specific errors; no answers persisted either way.		
Priority:	Must		
Frequency of Use:	Low to medium		
Flow of Events:	1. Visitor opens the indicative HFE pre-check. 2. System displays the indicative-use disclosure and exclusions. 3. Visitor acknowledges the disclosure and completes the guided questionnaire. 4. System validates the session-only answers. 5. System evaluates the answers using the versioned ruleset. 6. System displays the classification, reasons, limitations, official sources, and ruleset date. 7. Visitor follows the official HDB link or closes the check; system discards the answers.		
Alternative Flows:	7.1.AC.1: Visitor does not acknowledge the disclaimer -> pre-check does not start; explanatory screen shown. 7.1.AC.2: Omitted required answer -> field-specific error; Visitor returns to that step. 7.1.AC.3: Invalid range value (e.g. age, income) -> field-specific error naming the acceptable range.		
Exceptions:	7.1.EX.1: Ruleset metadata unavailable -> pre-check halted with an explanatory error; no classification produced.		
Includes:	None.		
Special Requirements:	Functional requirements: FR-HFE-01 to FR-HFE-10. Additional constraints: NFR-PRIV-01 (zero data persistence; session data discarded on completion/timeout).		
Assumptions:	HFE ruleset is team-authored from public guidelines and provides indicative guidance only		
Notes and Issues:	None.		

Use Case 8.1 View Data Source Sync Status

Use Case ID:	8.1		
Use Case Name:	View Data Source Sync Status		
Created By:	Banana Five-O	Last Updated By:	Banana Five-O
Date Created:	8 September 2026	Date Last Updated:	8 September 2026
Actor:	Data Administrator		
Description:	Displays each configured data source's synchronisation status to a Data Administrator. Trigger: the Data Administrator opens the source-synchronisation dashboard.		
Preconditions:	Data Administrator is authenticated and holds the Data Administrator role.		
Postconditions:	Success: Source-status summary displayed. Failure: Access denied for non-administrators.		
Priority:	Must		
Frequency of Use:	Medium		
Flow of Events:	1. Data Administrator navigates to the administration page. 2. System verifies the Data Administrator role. 3. System retrieves active source status, last sync timestamp, record count, and reverse-chronological sync history (start/end time, outcome, records changed, sanitized error message). 4. System displays the source-status dashboard and filterable synchronization history table.		
Alternative Flows:	8.1.AC.1: Source has never run -> "never run" status shown. 8.1.AC.2: Last run failed -> failure status shown, linked to synchronisation history (see FR-ADM-05).		
Exceptions:	8.1.EX.1: Anonymous or non-Data Administrator access -> access denied at both UI and API levels.		
Includes:	None.		
Special Requirements:	Functional requirements: FR-ADM-01, FR-ADM-02, FR-ADM-05. Additional constraints: NFR-SEC-03 (non-administrators cannot read or invoke admin functions, even by direct endpoint call).		
Assumptions:	None.		
Notes and Issues:	None.		

Use Case 8.2 Trigger Manual Source Refresh

Use Case ID:	8.2		
Use Case Name:	Trigger Manual Source Refresh		
Created By:	Banana Five-O	Last Updated By:	Banana Five-O
Date Created:	8 September 2026	Date Last Updated:	8 September 2026
Actor:	Data Administrator		
Description:	Allows a Data Administrator to request a manual refresh for one configured data source at a time. Trigger: the Data Administrator selects one configured source and requests a refresh.		
Preconditions:	1. Data Administrator is authenticated and authorised. 2. At least one configured source is selectable.		
Postconditions:	Success: Refresh job accepted and an identifier returned. Failure: Rejection reason shown; no job started.		
Priority:	Must		
Frequency of Use:	Low		
Flow of Events:	1. Data Administrator opens the manual-refresh control. 2. Data Administrator selects one configured source. 3. Data Administrator requests a refresh. 4. System verifies the administrator role and confirms no refresh is already running for that source. 5. System accepts the request and creates a refresh job. 6. System displays the accepted job identifier. 7. System records the completed outcome in synchronisation history.		
Alternative Flows:	8.2.AC.1: Unknown source -> rejected with a reason. 8.2.AC.2: Concurrent request for the same source -> rejected with a reason ("A refresh is already running for this source"). 8.2.AC.3: Unauthorised request -> rejected.		
Exceptions:	8.2.EX.1: Refresh fails validation mid-run -> previous validated dataset preserved (see 8.3); failure recorded in history.		
Includes:	None.		
Special Requirements:	Functional requirements: FR-ADM-01, FR-ADM-03. Additional constraints: NFR-SEC-03; C-4 (external providers called within published rate limits).		
Assumptions:	Only one source may be refreshed at a time per request.		
Notes and Issues:	None.		

Use Case 8.3 Run Scheduled Daily Sync

Use Case ID:	8.3		
Use Case Name:	Run Scheduled Daily Sync		
Created By:	Banana Five-O	Last Updated By:	Banana Five-O
Date Created:	8 September 2026	Date Last Updated:	8 September 2026
Actor:	System (scheduled process)		
Description:	Automatically checks each configured source for an available update at least once every 24 hours, and preserves the last validated dataset if a refresh fails. Trigger: The scheduled process reaches the configured update-check time for a source.		
Preconditions:	A scheduled trigger fires.		
Postconditions:	Success: Update check recorded; refresh executed if an update was available and validated successfully. Failure: Previous validated dataset remains queryable; failure recorded.		
Priority:	Must		
Frequency of Use:	Automatic (at least every 24 hours per source)		
Flow of Events:	1. The scheduled process triggers an update check for a configured source. 2. System checks the source and detects an available update. 3. System downloads the candidate data within the provider's rate limits. 4. System validates the candidate schema and content. 5. System activates the validated dataset version. 6. System records the check, decision, record counts, and successful outcome. 7. System makes the new dataset available to dependent features.		
Alternative Flows:	8.3.AC.1: No update is available; the system records the check and retains the active dataset.		
Exceptions:	8.3.EX.1: Download, schema, or validation failure; the system preserves the last validated dataset, keeps it queryable, and records a sanitised failure.		
Includes:	None.		
Special Requirements:	Functional requirements: FR-ADM-04, FR-RES-01. Additional constraints: C-2 (transaction store scales to $\geq 1,000,000$ rows with indexed queries); C-4 (rate limits respected).		
Assumptions:	A validated dataset from a prior run is always available as a fallback after the first successful sync.		
Notes and Issues:	None.		